

Evan Gossling

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Education

Doctor of Philosophy (Ph.D.) in Computer Engineering 2023 – Present
Iowa State University of Science and Technology GPA: 4.00/4.00

Bachelor of Science (B.S.) in Computer Engineering 2019 – 2023
Iowa State University of Science and Technology GPA: 3.85/4.00

Experience

Graduate Research Assistant, Iowa State University – Ames, IA Aug. 2023 – Present

- ❖ Researching low-latency, broadband transport for rural 5G and Non-Terrestrial Networks (NTN) under Dr. Hongwei Zhang
- ❖ Conducting research for Collins Aerospace on multipath packet scheduling, NTN, and network path modeling
- ❖ Developing MPQUIC algorithms and protocols for high-performance multipath transport
 - Investigating intelligent multipath scheduling for Satellite–Terrestrial Integrated Networks (STINs) supporting real-time applications
 - Combining MPQUIC, intelligent packet scheduling, erasure coding (FEC), and per-path network models into a holistic transport solution
 - (Ongoing) Demonstrating improved real-time video performance, including increased FPS and 50% reductions in frame and packet loss
- ❖ Researching an end-to-end RaptorQ erasure coding framework with BitRipple, Inc. and Boston University for low-latency, broadband applications in rural wireless environments
- ❖ Integrating RaptorQ into transport protocols for reliable packet delivery over dynamic and lossy wireless networks
 - Evaluated Starlink NTN transfers using Cubic and BBR; achieved 10 Mbps versus 1–2 Mbps with Cubic alone and 95% fewer retransmissions with BBR
 - Evaluated rural 5G teleoperation; achieved stable target-rate FPS, up to 3× fewer missed-frame events than Reed-Solomon and 56× fewer than baseline, with 50–69% smoother frame delivery

Undergraduate Research Assistant, Iowa State University – Ames, IA Sep. 2022 – Aug. 2023

- ❖ Researched networking for 5G and beyond networks under Dr. Hongwei Zhang with the WiCi Center and \$16M ARA PAWR Platform
- ❖ Collaborated on deployment of the ARA Platform for wireless technology research across 5G, Beyond-5G, and NTN networks
- ❖ Deployed Power Distribution Units (PDUs) at 5G base stations and developed remote monitoring and management software
- ❖ Developed a containerized API and SQL database to monitor PDU status and power draw, remotely control power usage, and collect real-time data

App Dev Intern, RSM US LLP – Des Moines, IA Jun. 2022 – Aug. 2022

- ❖ Developed internal client software within the consulting division of a leading tax, audit, and consulting firm
- ❖ Applied object-oriented programming to develop high-performance scripts, including reducing legacy script loading times
- ❖ Presented solutions to clients, incorporated feedback, and iterated to meet client requirements

Undergraduate Research Assistant, Iowa State University – Ames, IA Jan. 2020 – Jan. 2022

- ❖ Researched geometric computing for smart urban and rural environments under Dr. Goce Trajcevski
- ❖ Collaborated on a web application visualizing compression of evolving Triangulated Irregular Networks

- ❖ Developed the front end using HTML, JavaScript, jQuery, Ajax, and D3.js
- ❖ Developed web applications using Flask and Django with a PostgreSQL backend

Skills

Programming Languages: Rust, Java, Python, C

Networking: MPQUIC, QUIC, TCP/IP, Linux networking, Wireshark

Web Development: Flask, HTML, JavaScript, CSS, jQuery

Tools/Technologies: Git, Google Cloud Platform, SQL, PostgreSQL

Virtualization/Containerization: Docker, VMware, VirtualBox

Awards & Honors

- ❖ National Science Foundation Graduate Research Fellowship (**NSF GRFP**) Apr. 2024
- ❖ Glenn W. and Cordelia R. Sellers Graduate Scholarship Aug. 2023
- ❖ Iowa State University Honors Program Aug. 2019 - Jan. 2023
- ❖ Best Demo Award, AgWireless'26, 2026
- ❖ Best Paper Honorable Mention, IEEE Future Networks World Forum (FNWF), 2023
- ❖ Best Paper Award, 17th ACM Workshop on Wireless Network Testbeds, Experimental evaluation and Characterization (WiNTECH), 2023

Organizations

- ❖ IEEE-Eta Kappa Nu Inducted Apr. 2021
 - ◆ President - Nu Chapter Aug. 2022 - Jul. 2026
 - ◆ Vice President - Nu Chapter Aug. 2021 - Aug. 2022
- ❖ IEEE (Student Member) Apr. 2021 - Present
 - ◆ IEEE Communications Society Aug. 2025 - Present
- ❖ ACM (Professional Member) Dec. 2023 - Present
- ❖ Tau Beta Pi (Member) Inducted Apr. 2021

Publications

Journal

- ❖ Taimoor Ul Islam, Joshua Ofori Boateng, Md Nadim, Guoying Zu, Mukaram Shahid, Xun Li, Tianyi Zhang, Salil Reddy, Wei Xu, Ataberk Atalar, Vincent Lee, Yung-Fu Chen, Evan Gossling, Elisabeth Permatasari, Christ Somiah, Owen Perrin, Zhibo Meng, Reshal Afzal, Sarath Babu, Mohammed Soliman, Ali Hussain, Daji Qiao, Mai Zheng, Ozdal Boyraz, Yong Guan, Anish Arora, Mohamed Selim, Arsalan Ahmad, Myra B. Cohen, Mike Luby, Ranveer Chandra, James Gross, Kate Keahey, Hongwei Zhang, Design and Implementation of ARA Wireless Living Lab for Rural Broadband and Applications, Computer Networks (2025).

Conference

- ❖ Owen Perrin, Evan Gossling, Daji Qiao, Hongwei Zhang. A Measurement Study of Dynamics and Liquid Data Transport in OneWeb LEO Satellite Networks. 2025 IEEE Military Communications Conference (MILCOM '25).
- ❖ E. K. A. Permatasari, E. Gossling, M. Nadim, S. Babu, D. Qiao, H. Zhang, M. Luby, J. W. Byers, L. Minder, and P. Aggrawal. 2024. Real-Time Liquid Wireless Transport for Video Streaming in Rural and Agricultural Applications. In Proceedings of the 3rd Mile-High Video Conference (MHV '24).
- ❖ (**Best Paper Honorable Mention**) Tianyi Zhang, Guoying Zu, Taimoor Ul Islam, Evan Gossling, Sarath Babu, Daji Qiao, and Hongwei Zhang. Exploring Wireless Channels in Rural Areas: A Comprehensive Measurement Study. 2023 IEEE Future Networks World Forum (FNWF '23).
- ❖ (**Best Paper Award**) Taimoor Ul Islam, Tianyi Zhang, Joshua Ofori Boateng, Evan Gossling, Guoying Zu, Sarath Babu, Hongwei Zhang, and Daji Qiao. 2023. AraMIMO: Programmable TVWS mMIMO Living Lab for

Rural Wireless. In Proceedings of the 17th ACM Workshop on Wireless Network Testbeds, Experimental evaluation & Characterization (WiNTECH '23).

Demos | Posters

- ❖ **(Best Demo Award)** Evan Gossling, Jalal Jalali, Daji Qiao, Hongwei Zhang, and Naveen Rajanikantha. 2026. Prediction Aware Multipath Scheduling in Satellite–Terrestrial Integrated Networks for Real-Time Applications (Demo). AgWireless'26.
- ❖ Evan Gossling, Elisabeth Permatasari, Md Nadim, Sarath Babu, Daji Qiao, Hongwei Zhang, Mike Luby, Lorenz Minder, Pooja Aggrawal, and John Byers. 2024. Real-Time Liquid Wireless Networking for Data-Intensive Rural Applications (Poster). 2024 Farm Progress Show, AraFest 2024.
- ❖ Mike Luby, Elisabeth Permatasari, and Evan Gossling. 2023. Real-Time Liquid Wireless Networking for Data Applications (Demo). 2023 ARA Public Launch.
- ❖ Ce Li, Will Postler, Ian Johnson, Paul Brinkmann, Evan Gossling, Bailey Gorlewski, and Goce Trajcevski. 2023. Geo-Awareness of Learnt Citations Prediction for Scientific Publications (Demo Paper). In Proceedings of the 7th ACM SIGSPATIAL Workshop on Location-based Recommendations, Geosocial Networks and Geoadvertising (LocalRec '23).
- ❖ Prabin Giri, Hooman Hashemi, Evan Gossling, Jason T Guo, Koshal P Shah, and Goce Trajcevski. 2020. CET-LATS: Compressing Evolution of TINs from Location Aware Time Series (Demo Paper). In Proceedings of the 28th International Conference on Advances in Geographic Information Systems (SIGSPATIAL '20).